

model_comparison_table.md

DreamCoder Model Versions: Comprehensive Comparison

Created: December 4, 2024 **Purpose:** Establish a canonical reference for all architectural variations in the DreamCoder codebase

1. RECOGNITION MODEL ARCHITECTURES

Component	Current GRU	Legacy GRU
File	neural_recognition.py	neural_recognition_legacy_gru.py
Encoder Type	Bidirectional GRU	Bidirectional GRU
Hand Encoding	GRU over per-card MLPs	GRU over per-card MLPs
Example Encoding	Concatenate hand + output	Concatenate hand + output
Task Encoding	Attention-weighted pooling	Mean pooling
Card Feature Dim	24	24
Hidden Dim / d_model	128 (default)	128 (default)
LayerNorm Usage	Post-attention & post-FF	Post-attention & post-FF
Positional Encoding	None	None
num_heads	N/A	N/A
num_sab_layers	N/A	N/A

Component	Current GRU	Legacy GRU
num_seeds (PMA)	N/A	N/A
Dropout	0.1	0.1
Optimizer	Adam (lr=1e-3)	Adam (lr=1e-3)
LR Scheduler	None	None
Gradient Clipping	None	None
Max Examples	20	20
Max Cards	8	8

Key Architectural Innovations

Set Transformer Task Encoding (Critical Fix - Dec 3, 2024):

- Problem: Learned embeddings ($d_{\text{model}}=64$) collapse discriminative information
- Solution: Compute correlations on **raw card features** (24-dim), not learned embeddings
- Method:
 - i. $\text{raw_correlation} = \text{corr}(\text{raw_card_features}, \text{labels})$ (24 dims)
 - ii. $\text{raw_diff} = \text{mean}(\text{pos_examples}) - \text{mean}(\text{neg_examples})$ (24 dims)
 - iii. Concatenate and expand to d_{model} via identity-initialized projection
- Result: Task embedding cosine similarity range changed from $[0.98, 1.0]$ to $[-0.38, 0.49]$

FiLM Modulation for Example Encoding:

- Problem: Concatenation of hand + label washes out label signal
- Solution: $\text{output} = \gamma * \text{hand_emb} + \beta$ where $\gamma, \beta = f(\text{label})$
- Ensures True/False labels produce fundamentally different representations

2. EXAMPLE GENERATION

Parameter	Random Sampling	Balanced Sampling (Fixed)
Method	<code>sample_hand(hand_size)</code>	Rejection sampling until 50/50
Training Examples	100	100 (50 positive, 50 negative)
Holdout Examples	20	20 (10 positive, 10 negative)
Hand Size	5	5
Problem	Extreme class imbalance for some rules	Fixed
Rules Skipped	None	3 (can't balance: <code>Halves_AP_len3_any_equal</code> , <code>Even_opens_next_closes</code> , <code>Odd_opens_next_closes</code>)

Base Rate Analysis (Random 5-card Hands)

Category	Count	Examples
>95% True	1	<code>Halves_AP_len3_any_equal</code> (100% True)
>95% False	18	<code>Even_opens_next_closes</code> , <code>Ranks_palindrome</code> , <code>Shift_half_plus_two</code> , etc.
Balanced (5-95%)	38	<code>Has_pair_ranks</code> (58.5%), <code>Ends_same_color</code> (58%), etc.

3. ENUMERATION PARAMETERS

Parameter	Default	Cython	Notes
Algorithm	Best-first search	Best-first search	Grammar-weighted by log probability
Max Depth	Phase-dependent (6-15)	6-15	Increases with phase

Parameter	Default	Cython	Notes
Max Programs	Phase-dependent (5K-2M)	Phase-dependent	Budget per task
Timeout per Task	60-600 sec	60-600 sec	Increases with phase
Parallel Workers	4	4	PyPy subprocesses
Early Pruning	All-or-nothing mode	Enabled	Skip partial solutions
Speedup vs Baseline	1x	~7-15x (theoretical)	Actual ~1.5x due to pickle serialization

4. WAKE-SLEEP CYCLE

Component	Implementation
Wake: Enumeration	Enumerate programs for each task, keep top-k solutions per frontier
Sleep: Compression	Anti-unification to find common patterns, add as invented primitives
Sleep: Recognition	Train neural model on solved tasks for N epochs
Sleep: Dreaming	Generate synthetic tasks from recognition model (optional)
Grammar Update	Reweight primitives based on solution frequencies
Library Growth	Typically 0-5 new abstractions per compression

5. PRIMITIVES COMPARISON

Aspect	card_primitives.py	lean_primitives.py	extended_primitives.py
Total Count	~61	~50	~70+
Philosophy	Comprehensive baseline	Cognitive realism	Academic rigor

Aspect	card_primitives.py	lean_primitives.py	extended_primitives.py
Constants	0-6, suits, colors	0-5, 10-14, 17, 21	Base lean + extended
Abstract Combinators	Yes (compose, flip, const, id)	No (low cognitive reality)	Yes
List Construction	Yes (cons, nil)	No (not how humans think)	Yes
Game-Relevant Numbers	No	Yes (10-14 for face cards, 17/21 for Blackjack)	No

Lean Primitives Philosophy

- Primitives should be "directly nameable" in short phrases
- Removed: compose , flip , const , id , cons , nil
- Added: 10 , 11 , 12 , 13 , 14 (face card values), 17 , 21 (game thresholds)
- Target: ~50 primitives with high cognitive realism

New Primitives (Dec 4, 2024)

- half_len : len(xs) // 2
- first_half : xs[:len(xs) // 2]
- second_half : xs[len(xs) // 2:]

6. RUNNER CONFIGURATIONS

run_overnight_set_transformer.py

Phase	Name	Iterations	Max Depth	Max Programs	Timeout	Recognition Epochs
1	Quick Exploration	5	6	5,000	60s	5
2	Medium Exploration	5	6	10,000	90s	5

Phase	Name	Iterations	Max Depth	Max Programs	Timeout	Recognition Epochs
3	Deep Search	10	7	50,000	120s	12
4	Intensive Search	10	8	100,000	180s	15

run_experimental_rules.py

Phase	Name	Iterations	Max Depth	Max Programs	Timeout	Recognition Epochs
1	Quick Exploration	5	7	100,000	60s	5
2	Medium Search	10	9	250,000	120s	8
3	Deep Search	15	11	500,000	180s	10
4	Extended Deep Search	20	13	1,000,000	300s	10
5	Exhaustive Search	15	15	2,000,000	600s	15

run_overnight_v4.py (Part A: Standard Batch)

Phase	Rule Set	Max Depth	Iterations	Recognition Epochs
1-2	Easy pretraining (level 1)	6	5 each	5
3-4	All pretraining	7-8	10 each	12-15
5-7	Experimental rules (transfer)	8-10	10-15 each	15-25

7. PARAMETER COMPARISON BY RUNNER

Parameter	v3	v4 A	v4 B	Set Transformer	Experimental
Recognition Model	GRU	GRU	GRU	Set Transformer	Set Transformer
Primitives	lean	lean	lean	lean	lean
Total Phases	4	7	7	4	5
Total Iterations	30+	60+	60+	30	65
Max Initial Depth	6	6	6	6	7
Max Final Depth	10	10	10	8	15
Workers	4	4	4	4	4
Mini-batch Compression	No	No	Yes	No	No
Balanced Sampling	No	No	No	No	Yes (fixed)

8. KEY FIXES AND LESSONS LEARNED

Task-Result Scrambling Bug (v3 Fix)

- **Problem:** `as_completed()` returns futures in arbitrary order, scrambling task-result mappings
- **Solution:** Use dictionary keyed by `task.name` instead of list index

Task Embedding Collapse (Set Transformer Fix - Dec 3, 2024)

- **Problem:** All task embeddings had cosine similarity ~ 1.0 (no discrimination)
- **Root Cause:** Neural network layers destroy discriminative information present in raw features
- **Solution:** Compute correlations on raw 24-dim card features, not learned embeddings
- **Result:** Cosine similarity range changed from $[0.98, 1.0]$ to $[-0.38, 0.49]$

Trivial Solutions Passing Verification (Dec 4, 2024 Fix)

- **Problem:** $(\lambda \text{ true})$ and $(\lambda \text{ false})$ passed holdout verification
- **Root Cause:** Random sampling produces extreme class imbalance for some rules
- **Solution:** Balanced sampling (rejection sampling for 50/50 split)
- **Result:** 54/57 rules can be balanced; 3 rules skipped (impossible to balance)

9. FILES REFERENCE

Component	Primary File	Notes
Current GRU Recognition	neural_recognition.py	Attention-weighted task encoding
Legacy GRU Recognition	neural_recognition_legacy_gru.py	Mean-pooled task encoding
Set Transformer Recognition	set_transformer_recognition.py	Raw correlation task encoding
DreamCoder Framework	dreamcoder_v2.py	Core Task, Frontier, Metrics classes
Lean Primitives	lean_primitives.py	Cognitive realism optimized
Extended Primitives	extended_primitives.py	Academic (Ellis et al.) style
Card Primitives	card_primitives.py	Legacy baseline
Compression	compression.py	Anti-unification
Enumeration	enumeration.py	Best-first search

10. CHECKLIST FOR NEW EXPERIMENTS

When running a new experiment, verify:

- ☐ **Example Sampling:** Balanced or random? (Use balanced for holdout verification)
- ☐ **Recognition Model:** GRU or Set Transformer?

- ☐ **Primitives:** lean, extended, or card?
- ☐ **Phase Configuration:** Depth, programs, timeout, epochs per phase
- ☐ **Rule Set:** Pretraining (43), experimental (57), or subset?
- ☐ **Workers:** 4 PyPy workers configured?
- ☐ **Checkpointing:** Per-iteration or per-phase?
- ☐ **Transfer:** Pretraining → experimental, or direct evaluation?

APPENDIX: Rule Base Rates (Random 5-Card Sampling)

Rules with extreme base rates (>95% or <5%) require balanced sampling:

Rule	Base Rate	Category
Halves_AP_len3_any_equal	100% True	Cannot balance
Even_opens_next_closes	0% True	Cannot balance
Odd_opens_next_closes	0% True	Cannot balance
Ranks_palindrome	0% True	Balanceable with effort
Shift_half_plus_two	0% True	Partially balanceable
Shift2_plus3	0% True	Partially balanceable
Uniform_color	4.5% True	Balanceable
Sorted_by_rank	2.5% True	Balanceable